

LUMAFLOW · REQUIREMENTS

LumaFlow requirements

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/lumaflow		
Source	docs/03-requirements.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.3	2026-09-25	Amish Chadha	Record Amish's decisions (LMF-DDR-001); R16 redefined to \$225; R7 design stress sourced (6.8 MPa); status from LMF-CAL-001
0.2	2026-09-25	Amish Chadha	First measurable requirements for TRL 2, with status against the concept estimates
0.1	2026-09-24	Amish Chadha	Initial scaffold

LumaFlow requirements

These are the requirements for the concept, checked by calculation at TRL 3 in LMF-CAL-001. The status column reports that note; bracketed tags such as [B5] point to the line of docs/04-calcs/sizing.py that carries the number. Three requirements are **not met**: R2 and R3 (dose) and R16 (cost). Four are at risk: R4, R5, R8 and R10. The targets were set as proposals at TRL 2; the dose targets (R2, R3) and the budget (R16) now follow Amish's decisions of 2026-09-25 in LMF-DDR-001.

Table 1. Requirements and status at TRL 3.

ID	Requirement	Target	Status (LMF-CAL-001)	Verification
R1	Deliver a useful drinking-water flow	2.0 L/min (0.5 gpm) at the faucet, capped by a flow restrictor	Met by design [A3]	Restrictor rating; flow calculation
R2	Deliver the Class A dose level in clear water	Reduction equivalent dose (RED) of 40 mJ/cm ² or more at 2.0 L/min and UV transmittance (UVT) of 90 %/cm	Not met (14.7 laminar to 19.0 plug mJ/cm ²) [B5]	Ray-trace dose calculation; later biodosimetry
R3	Deliver the Class A dose level at the NSF/ANSI 55 test water quality	RED of 40 mJ/cm ² or more at 2.0 L/min and 70 %/cm UVT	Not met (6.3 to 7.4 mJ/cm ²) [B6]; kept visibly not met by decision D1	Dose calculation
R4	Monitor the dose and fail safe	UV-C sensor reading through the water column; visual and audible alarm and outlet valve closed within 1 s when the estimated dose falls below 40 mJ/cm ² , the sensor fails or power is lost	At risk: the far-end signal is about 2.4 nA at 90 %/cm and unresolved at 70 %/cm, and while R2 is not met the alarm would trip in any water [C1, C3]	Signal calculation; later firmware and test
R5	Switch with the flow	LEDs on within 0.5 s of flow above 0.3 L/min; LEDs held on for 5 s after flow stops; LEDs off otherwise	At risk (0.46 s worst case) [D2]	Timing calculation; later firmware sketch review
R6	No mercury and no warm-up	No mercury in any part; full UV output within 0.1 s of switch-on	Met by design	Design review; LED datasheet
R7	Withstand household water pressure	Working pressure 8 bar (116 psi); quartz window stress 6.8 MPa (1,000 psi) or less at 8 bar	Met (4.46 MPa with a 6 mm window) [G4]	Plate stress calculation
R8	Low pressure drop	0.5 bar (7 psi) or less across the unit at 2.0 L/min	At risk (0.42 bar on assumed flow coefficients, calculated without the flow restrictor, which absorbs excess supply pressure by design; clarifying the wording is proposed, awaiting Amish) [G2]	Component data and hydraulic calculation
R9	Low energy use	25 W or less while water flows; 0.5 W or less on standby	Met (22.6 W and 0.21 W) [E1, E2]	Power budget calculation
R10	Keep the LEDs cool	LED board 50 °C or less with 30 °C cabinet air and 25 °C water, including the 5 s run-on	At risk (48.9 °C nominal, 44.1 to 55.9 °C over the film-coefficient range) [F3]	Thermal calculation
R11	Food-safe wetted parts	Every wetted material food-contact grade (316 stainless steel, PTFE, fused quartz, EPDM or silicone O-rings, acetal); no UV-C on any plastic other than PTFE	Met by design (316 lower cap; PTFE liner, top disc and 316 insert shield the acetal upper cap)	Material review of the model
R12	Contain the UV-C light	No UV-C outside the unit in normal use; LEDs cannot run with the LED head or enclosure lid removed	Met by design	Design review of interlock and light path

ID	Requirement	Target	Status (LMF-CAL-001)	Verification
R13	Electrical safety	Only 24 V DC at the unit; mains confined to a certified external adapter; fuse on the 24 V input	Met by design	Design review
R14	Fit under a kitchen sink	Envelope 350 x 150 x 350 mm (13.8 x 5.9 x 13.8 in) or less, excluding the adapter	Met (235 x 72 x 322 mm) [H1]	Parametric model
R15	Easy service	Window and LED head replaceable with hand tools in 15 min or less; status LED shows run, fault and service due	Not verifiable at TRL 3 (four tie rods and two push-fit ports; needs a build to time)	Timed service on a build
R16	Low cost and buildable	Parts cost \$225 or less per unit (redefined from \$200 by decision D2); no custom PCB for the first build	Not met (\$249.00) [I1]	Priced BOM

■ Assumptions

- **Water quality.** The design water is clear, pre-filtered well water (decision D9): UVT of 90 %/cm or better at 275 nm, turbidity below 1 NTU, after a 5 µm sediment pre-filter. UVT must be measured for each site. Iron, manganese and hardness foul the window and must be within the limits commonly given for UV systems; this is an installation requirement, not a LumaFlow feature.
- **Dose target.** 40 mJ/cm² is the NSF/ANSI 55 Class A dose level. Reaching it would not make LumaFlow Class A compliant or certified. It inactivates bacteria and protozoa such as *Cryptosporidium* by several logs but gives a much smaller log reduction of adenovirus, which the US EPA dose tables show needs about 186 mJ/cm² for 4 logs ([EPA UV toolkit](#)). RED in LMF-CAL-001 is computed for a challenge organism of 20 mJ/cm² per log.
- **Pressure.** A pressure-limiting valve upstream is an installation requirement where supply can exceed 6 bar or water hammer occurs (LMF-CAL-001, G5).
- **Demand.** 10 to 20 L per household per day, drawn in about 20 short draws.
- R2, R3 and R16 are kept as stated so that the gaps stay visible. The routes to close R2 and R16 are proposals awaiting Amish in `docs/REVIEW.md`.